Structured notes: Module III: semiconductors

Puri and Babbar, *Solid State Physics and Electronics*, chapter 7 (printed pages 199–227). Text extracted by Codex from the page images and checked against them. Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption. Chapter questions and problems are not included.

Section 1: semiconductors, intrinsic and extrinsic (7.1–7.3)

7.1–7.3 Semiconductors, intrinsic and extrinsic (structured notes)

Source: book-sources/bs-5-semiconductors.html, Puri and Babbar pp. 199–204. Equations 7.1 and 7.2 are cited by their book numbers. Figures are redrawn.

Before you start

- Resistivity ranges of conductors, semiconductors and insulators.
- Energy bands in solids: the valence band, the conduction band and the gap E_g.

> **PLUS TWO RECALL: BANDS AND THE ENERGY GAP**
>
> **Recall.** In a solid the allowed energies form bands. The occupied band of highest energy is the valence band; the next band is the conduction band; the gap between them is E_g.
>
> **Re-derive.**
>
> 1. A conductor has a partly filled top band, so electrons can move under a field.
> 2. An insulator has a filled valence band and a wide gap, so electrons cannot move.
> 3. A semiconductor has the same band structure as an insulator with a narrow gap, 0.2 to 2.5 eV (book p. 199).
>
> **Saved.** Book p. 199. Diamond, with a gap of about 6 eV, is an insulator; Ge and Si are semiconductors at room temperature.

7.1 Introduction

Semiconductors have conductivities between those of conductors and insulators. Their resistivity lies between 10^{-5} and 10^{+4} ohm-m. Elemental semiconductors (Ge, Si) come from Group IV; compound semiconductors (GaAs, InP, CdS) come from Groups III and V, or II and VI. The band gap is small (0.2 to 2.5 eV), which gives the wavelengths used in LEDs and lasers. The resistance falls with temperature, and the conductivity can be controlled by doping.

7.2 Pure or intrinsic semiconductors

An intrinsic semiconductor is undoped. In silicon, each atom forms four covalent bonds, so the crystal is an insulator at 0 K (Fig. 7.1a). Thermal breaking of a bond frees an electron and leaves a hole; both carry current (Fig. 7.1b). At 0 K the valence band is full and the conduction band is empty (Fig. 7.2). Thermal excitation populates the conduction band and leaves an equal number of holes in the valence band, so $n_i = p_i$.

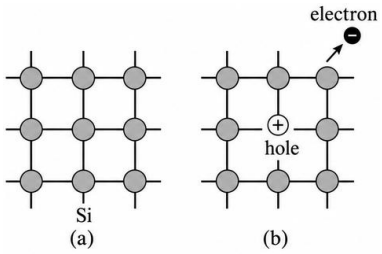


Fig. 7.1 (book): (a) two-dimensional representation of a Si crystal; (b) silicon crystal containing an electron-hole pair.

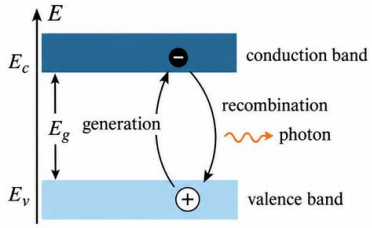


Fig. 7.2 (book): energy band diagram of an intrinsic semiconductor showing generation and recombination of an electron and a hole.

Holes have a larger effective mass and a lower mobility than electrons. Their charge is equal and opposite to the electron charge.

Recombination removes an electron-hole pair and releases the band gap energy as radiation. In equilibrium the generation rate g equals the recombination rate R. Since R depends on the carrier densities, $R = C\,n_i p_i$, and with $n_i = p_i$:

$$g = R = C\,n_i^2 \tag{7.1}$$

C depends on the material; it is related to the densities of states at the band edges (see §7.5).

7.3 Impurity or extrinsic semiconductors

Doping with Group V (donor) or Group III (acceptor) atoms, at about 1 in 10^6 to 10^8 atoms, changes the conductivity without changing the crystal structure. Donor atoms give excess electrons; acceptor atoms give excess holes.

7.3.1 Donor or n-type semiconductor

A pentavalent atom (P, As, Sb) forms four bonds, and its fifth electron is weakly bound (Fig. 7.3a). The donor level lies just below the conduction band, about 0.01 eV below in Ge and 0.03 eV below in Si, so the electron is easily promoted. The ionised donor is a fixed positive ion. Electrons are the majority carriers and holes the minority carriers. The product np is fixed (Eq. 7.2):

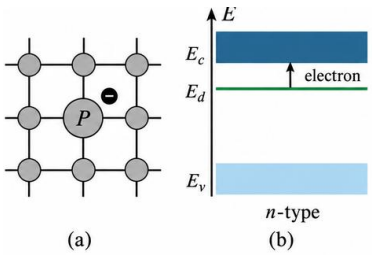


Fig. 7.3 (book): (a) a pentavalent impurity atom (P) in a silicon crystal; (b) energy level diagram of an n-type semiconductor.

$$np = n_i p_i = n_i^2 \tag{7.2}$$

Equation (7.2) is the law of mass action, derived in §7.5.4.

7.3.2 Acceptor or p-type semiconductor

A trivalent atom (B, Al, Ga, In) has one bond incomplete. It accepts an electron from a neighbour, which leaves a hole (Fig. 7.4a). The acceptor level lies about 0.01 eV above the valence band in Ge and 0.046 to 0.16 eV above it in Si. Holes are the majority carriers, electrons the minority carriers, and $np = n_i^2$ still holds. The crystal remains neutral, because the ionised acceptor is a fixed negative ion balanced by the mobile holes.

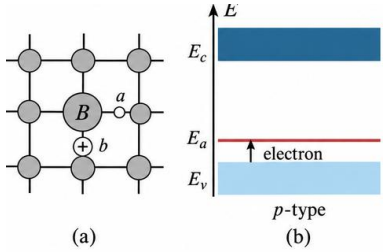


Fig. 7.4 (book): (a) a trivalent impurity atom (B) in a silicon crystal; (b) energy level diagram of a p-type semiconductor.

7.4 Drift velocity, mobility and conductivity of intrinsic semiconductors (structured notes)

Source: book-sources/bs-5-semiconductors.html, Puri and Babbar pp. 205–207. Equations 7.3 to 7.12 are cited by their book numbers. Figure is redrawn.

Before you start

- Current density J and Ohm's law $J = \sigma\mathcal{E}$.
- Intrinsic carrier concentration n_i (§7.2) and its temperature dependence (Eq. 7.10).

> **PLUS TWO RECALL: CURRENT DENSITY, DRIFT VELOCITY AND CONDUCTIVITY**
>
> **Recall.** $J = nev_d$ for carriers of density n, charge e and drift velocity v_d.
>
> **Re-derive.**
>
> 1. An electron accelerates in the field for average time τ, so $v_d = e\mathcal{E}\tau/m$.
> 2. $J = (ne^2\tau/m)\mathcal{E}$, so $\sigma = ne^2\tau/m$.
> 3. With $v_d = \mu\mathcal{E}$, $\mu = e\tau/m$ and $\sigma = ne\mu$ (Eq. 7.6).
>
> **Saved.** Eqs. (7.3), (7.5), (7.6).

7.4.1 Drift velocity and mobility

Thermal excitation keeps some carriers in both bands. With no field the motion is random and the net current is zero. Under a field each carrier gains velocity between collisions and loses it at each collision. The average extra velocity is the drift velocity v_d, which is proportional to the field:

$$v_d \propto \mathcal{E}$$

$$v_d = \mu\mathcal{E} \tag{7.3}$$

The mobility μ is the drift velocity per unit field. Electron and hole mobilities differ, because the two bands differ in occupancy.

7.4.2 Conductivity of the intrinsic semiconductor

For electrons, the current density is

$$J_n = nev_{dn} \tag{7.4}$$

With $v_{dn} = \mu_n\mathcal{E}$ this becomes $J_n = ne\mu_n\mathcal{E}$ (7.5). Ohm's law $J_n = \sigma_n\mathcal{E}$ then gives

$$\sigma_n = ne\mu_n \tag{7.6}$$

Holes give

$$\sigma_p = pe\mu_p \tag{7.7}$$

The total conductivity is

$$\begin{aligned}\sigma &= \sigma_n + \sigma_p \\ &= e\,(n\mu_n + p\mu_p)\end{aligned} \tag{7.8}$$

For an intrinsic semiconductor $n = p = n_i$, so

$$\sigma = en_i\,(\mu_n + \mu_p) \tag{7.9}$$

Drift current is due to the field; diffusion current is due to a concentration gradient. The treatment here is drift only, which is exact for uniform doping.

7.4.3 Temperature dependence of the conductivity

With mobilities independent of temperature, the variation of σ follows n_i, given by (7.10) (derived as Eq. 7.36):

$$n_i = \frac{2(2\pi kT)^{3/2}}{h^3}\,(m_n^* m_p^*)^{3/4} \exp\left(-\frac{E_g}{2kT}\right) \tag{7.10}$$

Substitution into (7.9) gives

$$\sigma = e(\mu_n + \mu_p)\,\frac{2(2\pi kT)^{3/2}}{h^3}\,(m_n^* m_p^*)^{3/4} \exp\left(-\frac{E_g}{2kT}\right) \tag{7.11}$$

Hence

$$\begin{aligned}\ln\sigma = &-\left(\frac{E_g}{2k}\right)\frac{1}{T} \\ &+ \frac{3}{2}\ln T \\ &+ \text{constant}\end{aligned} \tag{7.12}$$

The first term dominates. A plot of $\ln\sigma$ against $1/T$ is a straight line of slope $-E_g/2k$ (Fig. 7.5), which gives the band gap.

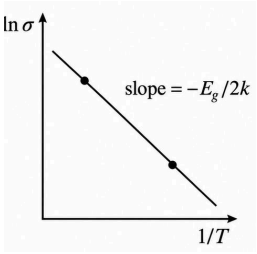


Fig. 7.5 (book): plot of ln σ against 1/T; the slope gives the band gap.

Section 3: carrier concentration and Fermi level, intrinsic (7.5)

7.5 Carrier concentration and Fermi level for intrinsic semiconductors (structured notes)

Source: book-sources/bs-5-semiconductors.html, Puri and Babbar pp. 207–213. Equations 7.13 to 7.36 are cited by their book numbers. Figure is redrawn. Unnumbered book equations are shown without a tag.

Before you start

- The Fermi–Dirac function (Eq. 7.14).
- Density of states for free electrons (Eq. 7.15).
- The Gamma-type integral $\int_0^\infty x^{1/2} e^{-x}\,dx = \sqrt{\pi}/2$.

The Plus Two base does not cover this section; no recall box is given.

7.5.1 Electron concentration in the conduction band

The electron density in $[E, E+dE]$ is

$$dn = D(E)\,f(E)\,dE \tag{7.13}$$

with the Fermi function (7.14) and the free-electron density of states (7.15). Combining (7.15) and (7.13) gives

$$dn = \frac{4\pi}{h^3}\,(2m)^{3/2} E^{1/2} f(E)\,dE \tag{7.16}$$

In a semiconductor the kinetic energy at E is $E - E_c$, so E is replaced by $E - E_c$ and the mass by m_n^*:

$$dn = \frac{4\pi}{h^3}\,(2m_n^*)^{3/2}(E-E_c)^{1/2}\,\frac{1}{\exp\left(\dfrac{E-E_F}{kT}\right)+1}\,dE \tag{7.17}$$

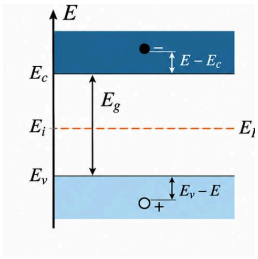


Fig. 7.6 (book): band model of an intrinsic semiconductor.

Integrate from E_c to infinity:

$$n = \frac{4\pi}{h^3}\,(2m_n^*)^{3/2}\int_{E_c}^{\infty}\frac{(E-E_c)^{1/2}\,dE}{\exp\left(\dfrac{E-E_F}{kT}\right)+1} \tag{7.18}$$

Near room temperature $kT \cong 0.026$ eV. For $E > E_c$ the 1 in the denominator is negligible. With $x = (E-E_c)/kT$ and the lower limit $x = 0$, the integral becomes $\int_0^\infty x^{1/2}e^{-x}dx = (\pi/4)^{1/2}$, and

$$n = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}\exp\left[-\left(\frac{E_c - E_F}{kT}\right)\right] \tag{7.19}$$

Write the prefactor as N_c, the effective density of states at the conduction band edge, so that $n = N_c\,f(E_c)$:

$$n = N_c \exp\left[-\left(\frac{E_c - E_F}{kT}\right)\right] \tag{7.20}$$

$$N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} \tag{7.21}$$

Silicon: $N_c = 2.8\times10^{25}(T/300)^{3/2}\ \mathrm{m^{-3}}$ (unnumbered in the book).

7.5.2 Hole concentration in the valence band

The density of holes is

$$dp = D(E)[1 - f(E)]\,dE \tag{7.22}$$

In the valence band $E < E_F$, so

$$1 - f(E) \cong \exp\left(\frac{E - E_F}{kT}\right) \tag{7.23}$$

A hole at E has kinetic energy $E_v - E$, so

$$D(E) = \frac{4\pi}{h^3}\,(2m_p^*)^{3/2}(E_v - E)^{1/2} \tag{7.24}$$

Integrating (7.22) from $-\infty$ to E_v, with $x = (E_v - E)/kT$, gives

$$p = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2} \exp\left[-\left(\frac{E_F - E_v}{kT}\right)\right] \tag{7.25}$$

$$p = N_v \exp\left[-\left(\frac{E_F - E_v}{kT}\right)\right] \tag{7.26}$$

$$N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2} \tag{7.27}$$

Silicon: $N_v = 2.8 \times 10^{25}(T/300)^{3/2}\ \mathrm{m^{-3}}$ (unnumbered). Equations (7.20) and (7.26) hold for intrinsic and extrinsic material. For intrinsic material, $E_F \to E_i$:

$$\begin{aligned} n_i &= N_c \exp\left[-\left(\frac{E_c - E_i}{kT}\right)\right], \quad p_i \\ &= N_v \exp\left[-\left(\frac{E_i - E_v}{kT}\right)\right] \end{aligned} \tag{7.28}$$

7.5.3 Fermi level

Setting $n = p = n_i$ in (7.20) and (7.26) and solving for E_F:

$$E_F = E_i = \frac{E_c + E_v}{2} + \frac{kT}{2}\ln\left(\frac{N_v}{N_c}\right) \tag{7.29}$$

$$E_F = \frac{E_c + E_v}{2} + \frac{3}{4}kT\ln\left(\frac{m_p^*}{m_n^*}\right) \tag{7.30}$$

At 0 K, $E_F = (E_c + E_v)/2$ (7.31): the Fermi level is midway in the gap. This holds at all T when $m_p^* = m_n^*$. For Si at 300 K the rise above the midgap value is about 0.01 eV.

$$E_F = \frac{E_c + E_v}{2} \tag{7.31}$$

7.5.4 Law of mass action and the intrinsic concentration

From (7.20) and (7.26) with $n = p = n_i$:

$$\begin{aligned} np &= n_i^2 \\ &= N_c N_v \exp\left(-\frac{E_c - E_v}{kT}\right) \\ &= N_c N_v \exp\left(-\frac{E_g}{kT}\right) \end{aligned} \tag{7.32}$$

With (7.21) and (7.27):

$$\begin{aligned} np &= n_i^2 \\ &= 4\left(\frac{2\pi kT}{h^2}\right)^3 (m_n^* m_p^*)^{3/2} e^{-E_g/kT} \end{aligned} \tag{7.33}$$

np depends only on T and the material. It equals n_i^2 in both intrinsic and extrinsic material (law of mass action):

$$np = n_i^2 \tag{7.34}$$

$$n_i = (N_c N_v)^{1/2} \exp\left(-\frac{E_g}{2kT}\right) \tag{7.35}$$

$$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2} (m_n^* m_p^*)^{3/4} \exp\left(-\frac{E_g}{2kT}\right) \tag{7.36}$$

Pure Ge at 300 K: $n_i \approx 2.4 \times 10^{19}\ \mathrm{m^{-3}}$ for 4.4×10^{28} atoms per cubic metre, about five bonds broken per 10^{10} atoms. Copper has about 10^{28} conduction electrons per cubic metre.

Section 4: extrinsic semiconductors, carriers, Fermi level and conductivity (7.6, Examples 7.1–7.3)

7.6 Carrier concentration, Fermi level and conductivity for extrinsic semiconductors (structured notes)

Source: book-sources/bs-5-semiconductors.html, Puri and Babbar pp. 213–223 (§7.6 and Examples 7.1 to 7.3). Equations 7.37 to 7.52 are cited by their book numbers; Eqs. 7.20 and 7.26 are reused. Figures are redrawn. The chapter summary, questions and problems are not included.

Before you start

- Donor and acceptor levels (§7.3); Fermi function (Eq. 7.14).
- Eqs. (7.20) and (7.26) for n and p; law of mass action (Eq. 7.34).
- Conductivity $\sigma = ne\mu$ (Eq. 7.6).

The Plus Two base does not cover this section; no recall box is given.

7.6(a) n-type semiconductor

At 0 K donor levels are occupied; with rising T, donors ionise. The Fermi level lies between E_d and E_c (Fig. 7.7).

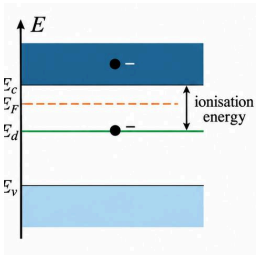


Fig. 7.7 (book): band model of an n-type semiconductor.

For E_F a few kT below E_c, $n = N_c \exp[-(E_c - E_F)/kT]$ (7.20). Charge balance:

$$n = N_d^+ + p \tag{7.37}$$

Neglecting thermal holes,

$$n = N_d^+ \tag{7.38}$$

so the minority carriers follow from $np = n_i^2$:

$$p = \frac{n_i^2}{N_d^+} \tag{7.39}$$

With

$$N_d^+ = N_d \exp\left[-\left(\frac{E_F - E_d}{kT}\right)\right] \tag{7.40}$$

taking logarithms gives

$$E_F = \frac{E_d + E_c}{2} + \frac{kT}{2}\ln\left(\frac{N_d}{N_c}\right) \tag{7.41}$$

Valid at moderate temperatures only. It fails at $T = 0$ K ($N_c \propto T^{3/2}$) and at very high T. As T rises, E_F falls through E_d to E_i (Fig. 7.8).

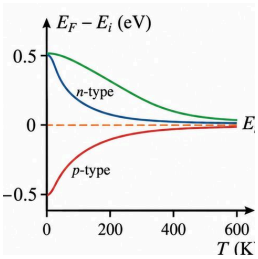


Fig. 7.8 (book): variations of the Fermi level with temperature and impurity concentration for Si.

$$n = n_i \exp\left(\frac{E_F - E_i}{kT}\right) \tag{7.42}$$

$$n = (N_d N_c)^{1/2} \exp\left(\frac{E_d - E_c}{2kT}\right) \tag{7.43}$$

$$n = (N_d N_c)^{1/2} \exp\left(-\frac{\Delta E}{2kT}\right) \tag{7.44}$$

$\Delta E = E_c - E_d$ is the donor ionisation energy. At moderate T, $n \propto \sqrt{N_d}$.

$$\sigma = en\mu_n \cong eN_d^+\mu_n \tag{7.45}$$

Fig. 7.9 (n-type Ge): (i) from about 10 K (A), σ rises as donors ionise, peaking at B (about 50 K) when all donors are ionised; (ii) from B to about room temperature (C), σ falls because the mobility falls; (iii) from C to D, σ rises sharply as the intrinsic conductivity grows faster than the mobility falls.

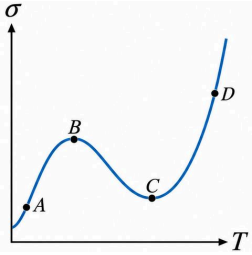


Fig. 7.9 (book): conductivity against temperature for a typical n-type germanium sample.

7.6(b) p-type semiconductor

Acceptors E_a lie above E_v. Some become ionised by taking valence electrons, creating holes. Charge balance $p = n + N_a^-$ with $n \ll N_a^-$ gives

$$p \cong N_a^- \tag{7.46}$$

where

$$N_a^- = \frac{N_a}{1 + \exp\left(\dfrac{E_a - E_F}{kT}\right)} \tag{7.47}$$

For $E_a - E_F \gg kT$,

$$N_a^- \cong N_a \exp\left[\frac{E_F - E_a}{kT}\right] \tag{7.48}$$

Equating with (7.26):

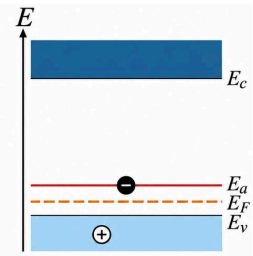


Fig. 7.10 (book): band model for a p-type semiconductor.

$$E_F = \frac{E_a + E_v}{2} - \frac{kT}{2}\ln\left(\frac{N_a}{N_v}\right) \tag{7.49}$$

Valid at moderate T. E_F rises with T towards E_i (Fig. 7.8). With (7.26) and (7.28):

$$p = n_i \exp\left(\frac{E_i - E_F}{kT}\right) \tag{7.50}$$

$$\sigma_p = ep\mu_p \cong eN_a^-\mu_p \tag{7.51}$$

The conductivity grows with either impurity concentration. For 10^{20} to 10^{22} m^{-3} dopant, $\rho \approx 10^{-3}$ to 10^{-1} ohm-m. Degenerate semiconductors (10^{23} to 10^{24} m^{-3}) approach metallic conductivity.

7.6(c) Mixed semiconductor

$$N_d^+ + p = N_a^- + n \tag{7.52}$$

For $N_d^+ = N_a^-$, $p = n = n_i$ (intrinsic behaviour). For $N_d \neq N_a$ the sample is n-type or p-type according to the larger concentration.

Example 7.1

Intrinsic Si at 300 K: $\mu_n = 0.135$, $\mu_p = 0.048$ m^2/V-s, $n_i = 1.5 \times 10^{16}$ m^{-3}.
$\sigma_i = en_i(\mu_n + \mu_p) = 4.39 \times 10^{-4}$ $(\Omega\text{-m})^{-1}$. After doping with 10^{23} P atoms per m^3: $n \cong 10^{23}$ m^{-3},
$p = n_i^2/n = 2.25 \times 10^9$ m^{-3},
$\sigma = en\mu_n = 21.6 \times 10^2$ $(\Omega\text{-m})^{-1}$, and
$E_F - E_i = kT\ln(n/n_i) = 0.406$ eV (Fig. 7.11).

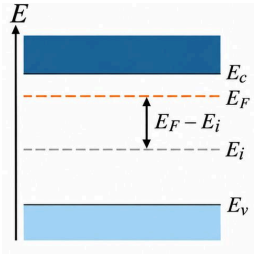


Fig. 7.11 (book): position of the Fermi level for Example 7.1 (the book prints the figure number only).

Example 7.2

Intrinsic GaAs: $\mu_n = 0.85$, $\mu_p = 0.04$ m^2/V-s; $m_n^* = 0.068\,m_0$, $m_p^* = 0.5\,m_0$; $E_g = 1.43$ eV at 300 K. From (7.36), $n_i = 1.94 \times 10^{12}$ m^{-3}. Then $\sigma = en_i(\mu_n + \mu_p) = 2.76 \times 10^{-7}$ $(\Omega\text{-m})^{-1}$.

Example 7.3

For intrinsic material, $\sigma = CT^{3/2}\exp(-E_g/2kT)$ with $C = 2e\mu(2\pi k/h^2)^{3/2}(m_n^* m_p^*)^{3/4}$. With $\sigma_1/\sigma_2 = (T_1/T_2)^{3/2}\exp\{(E_g/2k)(1/T_2 - 1/T_1)\}$, $T_1 = 293$ K, $\rho_1 = 4.5$ ohm-m, $T_2 = 305$ K, $\rho_2 = 2.0$ ohm-m, the book gives $E_g = 1.54 \times 10^{-19}$ J $= 0.96$ eV.